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**Cyber Solutions – The Intersection Of AI, Quantum And
Cybersecurity With Prof. Chuck Brooks**

CYBER SOLUTIONS – THE INTERSECTION OF AI, QUANTUM AND CYBERSECURITY WITH PROF. CHUCK BROOKS

April 13, 2026 @ 1:00 pm EDT

Feedback

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Brooks, a distinguished executive
to the forefront of technology.
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BROOKS

INSIDE CYBER

WILEY

HOW AI, 5G, IoT, AND QUANTUM COMPUTING WILL
TRANSFORM PRIVACY AND OUR SECURITY

INSIDE CYBER

CHUCK BROOKS

WILEY

Contemporary innovation is progressing so rapidly that it's not unusual to hear the pace of technological change being described by adjectives like "exponential" or "explosive." Indeed, it seems we are all witnessing a paradigm shift in the way societies deal with progress in applied scientific knowledge.

In *Inside Cyber: How AI, 5G, IoT, and Quantum Computing Will Transform Privacy and Our Security*, celebrated executive and consultant Chuck Brooks delivers an insightful exploration of the emerging technologies that will impact industry in the coming years and decades. Brooks explains the ramifications of artificial intelligence, machine learning, robotic process automation, predictive analytics, 5G, 3D printing, blockchain, quantum computing and other areas of tech.

The book elaborates on the challenges posed by the rapid expansion and proliferation of barely understood technologies and industries with the potential to disrupt core areas of commerce and business, as well as the opportunities presented by the same dramatic shift. It discusses the exposure of our most vital and critical infrastructures to these emerging technologies and talks about the network, cloud, and supply chain security issues that will arise for future businesses, both large and small.

(Continued on Back flap)

We're entering a rapidly evolving Fourth Industrial Era, where digital and physical worlds converge. The pace of technological advancement is so rapid that it's reshaping industries globally



Chuck Brooks

President

Brooks Consulting International

Technology is rapidly advancing, introducing both opportunities and significant cybersecurity challenges.

- ▶ **Rapid Evolution:** Technology is evolving at an unprecedented pace.
- ▶ **Interconnectedness:** Increasing interconnectedness of systems and devices.
- ▶ **New Threats:** Emerging technologies create new attack vectors and risks.



Technologies Impacting Digital Transformation

The world will be transformed by a seamless blend of digital and physical technologies



Artificial Intelligence (AI): is revolutionizing business via automation, predictive analytics, and tailored consumer experiences. AI will be the backbone of the economy and transformation.

Cloud Computing: Cloud technology offers the infrastructure for data storage, processing, and access, facilitating company scalability and the adoption of new technologies.

The Internet of Things (IoT): links devices and sensors to the internet, producing extensive data that may enhance operations, increase efficiency, and provide new income streams.

Quantum Computing: Has the potential to change the digital transformation process by providing capabilities that go far beyond those of traditional computers.

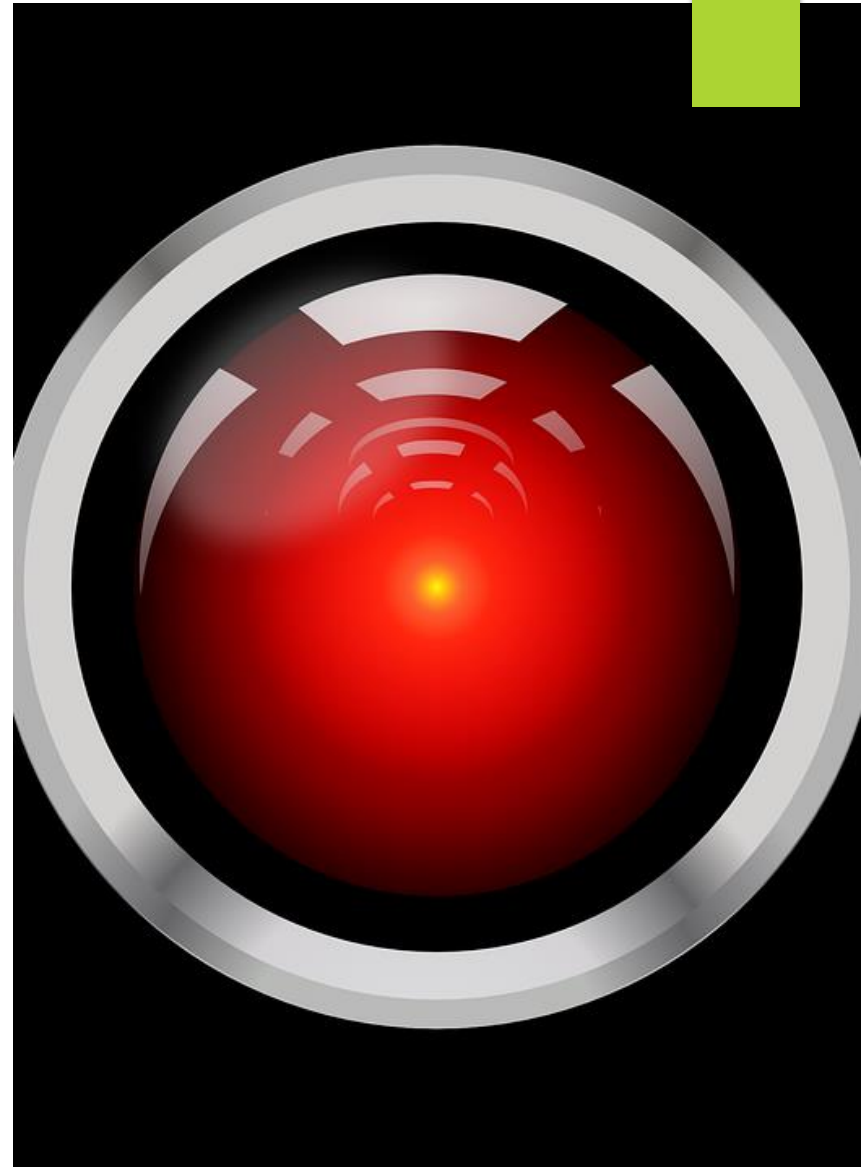
Blockchain Technology: provides safe and transparent mechanisms for the administration of transactions and data, especially in sectors such as supply chain management and banking.

Big Data Analytics: The examination of extensive information enables enterprises to discern patterns, comprehend consumer behavior, and make better educated choices.

5G: With the implementation of 5G and new satellites, communication will become more efficient, leading to a more seamlessly connected world.

Artificial Intelligence

- ▶ Artificial intelligence (AI) is an area of computer science that emphasizes the creation of intelligent machines that work and react like humans. Some of the activities computers with artificial intelligence are designed for include:
 - ▶ Speech recognition
 - ▶ Learning
 - ▶ Planning
 - ▶ Problem solving



The Transformational Impact of Artificial Intelligence



- **Automation and Efficiency:** AI is automating tasks, leading to increased efficiency and reduced costs in various processes, analytics, and activities across industries.
- **Enhanced Data Analysis:** AI's capabilities in ingesting, analyzing, and synthesizing data are revolutionizing fields like cybersecurity by enabling advanced threat prediction and real-time network monitoring.
- **Intersection with Emerging Technologies:** AI's effectiveness is magnified when combined with other technologies such as robotics, autonomous vehicles, and the Internet of Things (IoT), leading to a wider range of applications and impactful changes.
- **Reshaping Business Practices:** AI is fundamentally changing how businesses plan, design, and operate, influencing decision-making and operational strategies.
- **Modernizing Industries:** AI technologies like Machine Learning (ML), Deep Learning, Natural Language Processing (NLP), and speech recognition are modernizing various industries and applications.
- **Impact on Work and the Economy:** AI will reshape job roles and the economy. While many fear job displacement, AI will likely create new efficiencies and potentially new job categories, requiring a workforce that adapts to using AI tools effectively.
- **Evolution of Humanity:** AI has the potential to influence human evolution, including intelligence, relationships, and even our biological aspects, as explored in a podcast on theconversation.com.
- **Cybersecurity Challenges:** AI poses significant cybersecurity challenges, as sophisticated AI tools are being used by both criminals and nation-states to launch more effective and damaging cyberattacks, particularly targeting critical infrastructure like energy and telecommunications.
- **The Human Element:** While AI will handle many analytical and computational tasks, there is an importance of preserving and developing uniquely human skills, such as creativity, empathy, and critical thinking, that AI currently struggles to replicate.

Distinctions Between Generative AI and Agentic AI:



Generative AI

- **Focus:** *Primarily on creating new content based on training data.*
- **Function:** Acts like a digital creator, producing various forms of media such as text, images, code, and music.
- **Mode of Operation:** Reactive, responding to user instructions and prompts.
- **Strengths:** Excellent at tasks requiring content generation, creativity, and pattern mimicking within a defined domain.
- **Limitations:** May produce inaccurate or biased outputs and can struggle with complex tasks requiring reasoning or real-world interaction.
- **Examples:** ChatGPT for text generation, DALL-E for image creation.

Agentic AI

- **Focus:** *On autonomous action and task execution within an environment.*
- **Function:** Acts like a strategic mastermind or digital assistant, capable of making decisions and performing complex tasks independently.
- **Mode of Operation:** Autonomous and adaptive, interacting with the environment, tools, and potentially other agents to achieve goals.
- **Strengths:** Excelling at complex, multi-step tasks, planning, and adaptation in dynamic environments.
- **Limitations:** Potential for difficult-to-audit decisions and challenges in ensuring alignment with human values.
- **Examples:** Autonomous robots, self-driving cars, and chemistry-focused AI agents like ChemCrow.

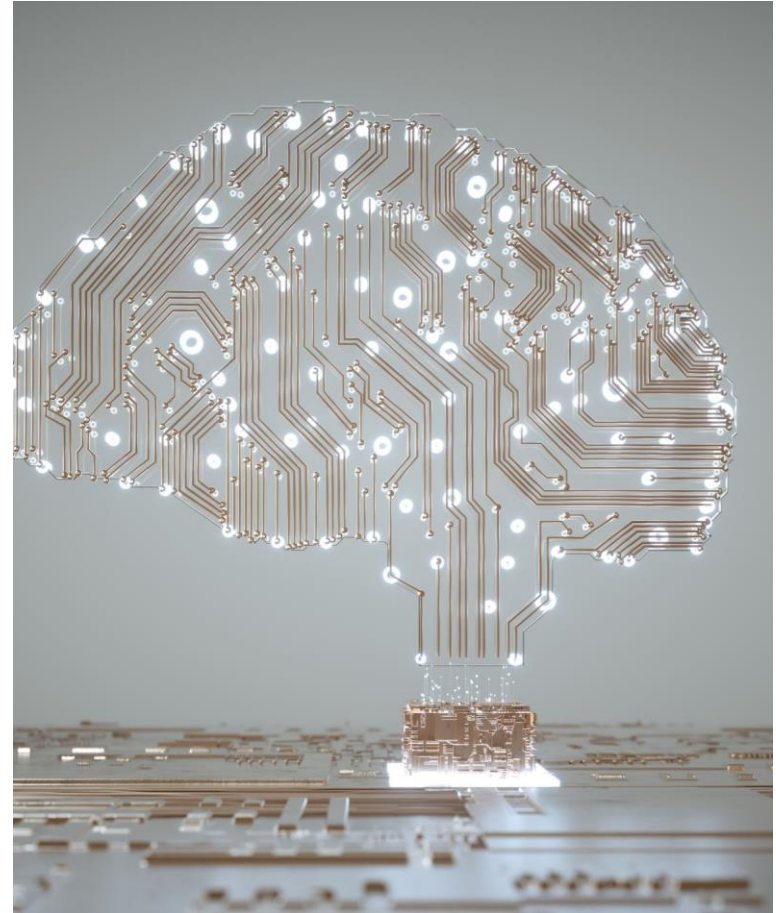
Cybersecurity & AI

- ▶ Technology is at an inflection point in history. AI and machine learning (ML) are advancing faster than society's ability to absorb and understand them.
- ▶ These new capabilities can make the world safer;; conversely, they introduce security challenges that could imperil public and private life.



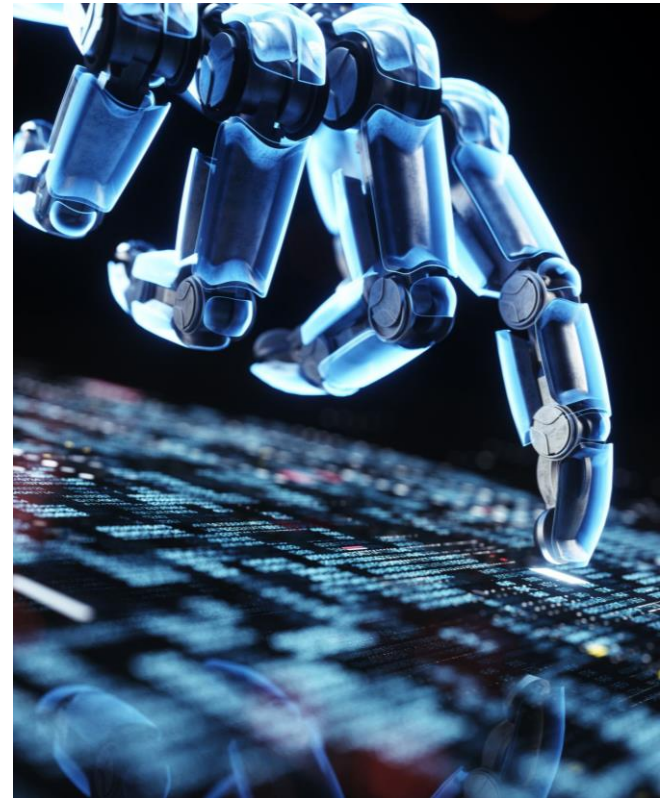
Uses Of Artificial Intelligence & Cybersecurity

- ▶ Threat Detection (Spam and Phishing)
- ▶ Malware Identification
- ▶ End-point detection
- ▶ Spam filtering and bot identification
- ▶ Password protection and user authentication
- ▶ Autonomous Patching
- ▶ Categorize Attacks. Lessen False/Positives
- ▶ Adapt To Evolving Risks – Predictive analytics
- ▶ Incident Response



Artificial Intelligence Uses In Cybersecurity

- ▶ AI as an enabler for cybersecurity has many applications
- ▶ AI's capabilities in predicting threats and synthesizing and correlating data is surely an advantage in mitigating threats. Especially in automation of process monitoring
- ▶ Automation allows for horizon scanning and monitoring of networks that can report on deviations and anomalies in real time. It allows for automatic updating of defense framework layers (network, payload, endpoint, firewalls and anti-virus) and diagnostic and forensics analysis for cybersecurity

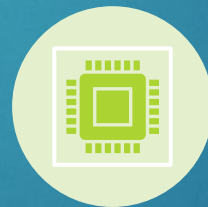


Other Artificial Intelligence Uses In Cybersecurity

INSIDER THREATS: AI can be used to mitigate insider threats, which are a significant challenge for organizations. By analyzing user behavior, AI-powered solutions can identify employees who may be engaging in malicious activities



FALSE POSITIVES: AI can help identify false positives, which is a major challenge for human analysts. This not only reduces the burden on human analysts but also makes threat detection and analysis more accurate and efficient



ACCESS CONTROL: AI can be used to strengthen access control measures. Machine learning algorithms can identify anomalous behavior patterns and flag suspicious login attempts, making it easier to identify potential security breaches.



Artificial Intelligence and Cybersecurity



Benefits of AI in cybersecurity

- **Automated Threat Detection and Response:** AI can significantly accelerate security processes by enabling real-time threat detection and analysis. It can effectively monitor networks for anomalies, identify emerging threats, and assist in evaluating network risks and vulnerabilities. This allows organizations to proactively address potential threats rather than solely reacting to incidents.
- **Enhanced Decision Making:** AI and machine learning can help organizations prioritize and act on data more efficiently, particularly in complex and extensive networks with many users and variables. This translates to better-informed and faster decision-making in the face of evolving threats.
- **Addressing Skills Gaps:** AI can help address the global shortage of cybersecurity experts and the increasing complexity of cybersecurity tools. AI-powered solutions can automate mundane tasks, freeing up human analysts to focus on higher-level strategic initiatives.
- **Real-time Analysis and Threat Identification:** AI enables continuous monitoring of network activity and helps identify unusual or suspicious actions that deviate from normal patterns, allowing for swift incident diagnosis and response.

Challenges and risks:

- **Dual-Edged Nature of AI:** While AI empowers defenders, it's also a powerful tool for adversaries. Cybercriminals can leverage AI for malicious purposes such as automating phishing attacks, creating more convincing spear-phishing messages using large language models, and developing sophisticated ransomware attacks.
- **Vulnerability in IoT and Critical Infrastructure:** The rise of the Internet of Things (IoT) and the increasing connectivity of devices create a larger attack surface, making securing critical infrastructure challenging.
- **Sophisticated Attacks:** Malicious actors and nation-states can use AI to identify vulnerabilities, develop new attack strategies, and target critical infrastructure with greater precision and speed.

The path forward:

- **Proactive Approach:** Companies and governments need to prioritize cybersecurity, invest in AI and machine learning capabilities, and take proactive measures to enhance their defenses.

A glowing green padlock is centered on the left side of the image. It is set against a dark background with a complex, glowing circuit board pattern. The padlock itself is a vibrant green with a textured, almost pixelated appearance, suggesting digital security or encryption.

Cybersecurity & AI- Challenges

- ▶ AI techniques could change the current asymmetric defender-versus-adversary balance in cybersecurity. The speed and accuracy of these advances will enable systems to act autonomously, to react and defend at wire speed, and to detect overt and covert adversarial reconnaissance and attacks.
- ▶ Therefore, securing the future requires substantial research investment in both AI and cybersecurity
- ▶ <https://www.nitrd.gov/news/2020/artificial-intelligence-cybersecurity.aspx>



Hackers Cybersecurity & AI

- ▶ As a tool for Hackers
- ▶ Cyber attackers use of offensive AI is becoming more sophisticated with the capability to self-mutate (polymorphic) as it learns from its environment while hiding in the background by leveraging existing data flows and human behavior to remain undetectable amid the noise across networks
- ▶ Malicious AI can also support intelligent malware learning from unsuccessful and successful attacks (Deep Exploit). Then smart malware can self-propagate by detecting and exploiting system vulnerabilities and adapt to mitigation measures by attempting new types of attacks

Living Machines: Will Artificial Intelligence Be Conscious? | IE (interestingengineering.com)

A view among experts that AI systems will probably reach overall human ability by 2040-50, and very likely by 2075



What Is Quantum Tech?



- ▶ Quantum computing uses the quantum state of an object to produce the basic unit of memory called a qubit.
- ▶ Unlike binary values that can either be 0 or 1, a qubit is not confined to a two-state solution, but can also exist in superposition. This means qubits can be employed at 0, 1, and both 1 and 0 simultaneously.
- ▶ This allows quantum computers to perform many calculations in parallel by pursuing simultaneous probabilities through superposition along with manipulating them with magnetic fields.
- ▶ Also, qubits can pair themselves via entanglement. Here, the state of one qubit cannot be described independently of the state of the others. So, a change in one qubit state will instantaneously change the state of the other, predictably.
- ▶ [All You Need to Know About Quantum Bits \(analyticsinsight.net\)](https://analyticsinsight.net)

Quantum Technologies



Quantum technology could revolutionize everything from genomic sequencing to transport route optimization, from code-breaking and decryption, from real-time analytics, to materials science.

Quantum computing could be directed and augmented via artificial intelligence, operate in a 5G or 6G framework, support IoT, and catalyze materials science, biotech, genomics, computer-brain interface, and the Metaverse

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Quantum technology could revolutionize everything from genomic sequencing to transport route optimization, from code-breaking and decryption, from real-time analytics to materials science.

But while quantum computers exist in the lab, general-purpose quantum computers are not yet available for commercial use.





Q-Day ?

- ▶ Large-scale quantum computers will be able to use Shor's algorithm to break all public key systems that employ RSA (integer factorization-based), Diffie—Hellman (finite field discrete log-based), and Elliptic Curve (elliptic curve discrete log-based) Cryptography. These algorithms underpin essentially all of the key exchange and digital signature systems in use today. Once reasonably sized quantum computers capable of operating on tens of thousands of logic quantum bits (qubits) exist, these public key algorithms will become useless.
- ▶ If Q-day happens it will require switching to stronger cryptographic systems, or cryptosystems. Decades of research in theoretical computer science has created post-quantum algorithms that seem impervious to cyber-attacks, even using mathematical approaches that consider quantum computing.

Quantum computing is poised to reshape society in profound ways —Here's a breakdown of its projected impacts:

⚡ Economic & Industrial Transformation

- **Boosting Innovation:** Industries like pharmaceuticals, finance, and logistics could accelerate discovery and optimization through quantum-enhanced simulations and data analysis.
 - **Competitive Advantage:** Early adopters may dominate markets by solving problems faster and more accurately, shifting global business dynamics.
- Job Evolution:** New roles in quantum software, hardware engineering, and ethics will emerge—while some existing roles may be automated or redefined.

🔒 Security & Privacy Challenges

- **Encryption Disruption:** Algorithms like Shor's threaten current cryptographic systems, requiring a shift to **post-quantum cryptography** to safeguard sensitive data.
- **New Threat Vectors:** Quantum capabilities may enable novel forms of cyber attacks if safeguards aren't rapidly developed and deployed.
- **Surveillance Potential:** Enhanced computational power may amplify data analysis in intelligence and law enforcement, raising privacy concerns.



🧠 Scientific Discovery & Problem Solving

- **Complex Simulations:** Quantum computing will help model molecular interactions, climate systems, and materials at atomic scales—unlocking deeper understanding of nature.

⚖️ Ethical & Societal Implications

- **Access Disparities:** Countries and corporations with quantum tech could widen digital divides—raising concerns about equity and control.
- **Regulation & Governance:** Policies will need to address ethical use, security standards, and international collaboration to prevent misuse.
- **Cultural Shifts:** Much like the internet's rise, quantum computing may change how we think about data, knowledge, and trust in technology.

RISKS TO SPACE BASED SYSTEMS

► Space is an emerging commercial critical infrastructure sector that is no longer the domain of only national government authorities. Space is an inherently risky environment in which to operate, so cybersecurity risks involving commercial space – including those affecting commercial satellite vehicles – need to be understood and managed alongside other types of risks to ensure safe and successful operations.

► -- NISTIR 8270



CYBER THREATS TO SPACE SYSTEMS

SPACE SEGMENT

- * Command Intrusion
- * Payload Control
- * Denial of Service
- * Malware

USER SEGMENT

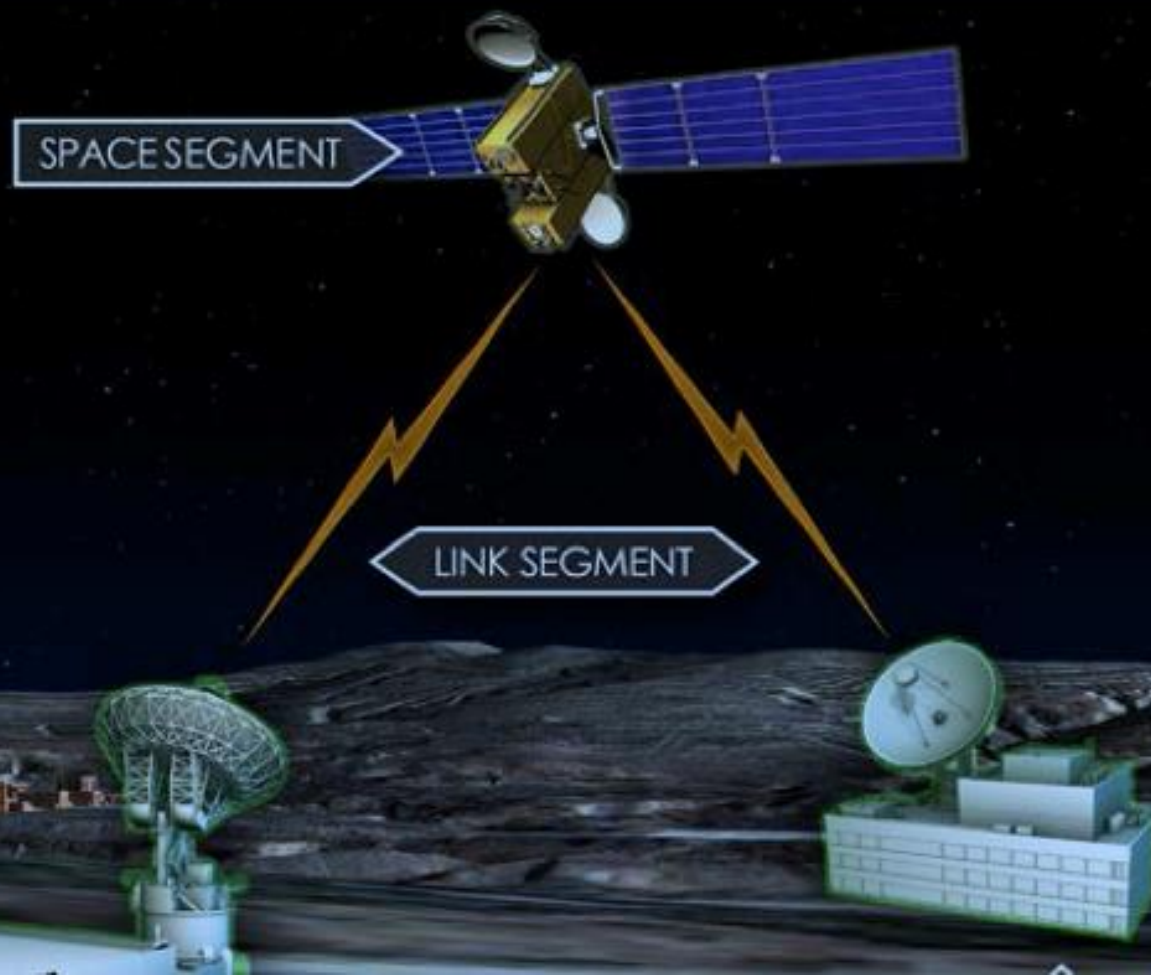
- * Spoofing
- * Denial of Service
- * Malware

LINK SEGMENT

- * Command Intrusion
- * Spoofing
- * Replay

GROUND SEGMENT

- * Hacking
- * Hijacking
- * Malware



CYBER VULNERABILITIES OF SATELLITES

- ▶ Satellite operations via Earth-bound entry points offer cyber attackers with an enormous number of potential inroads for hacking.
- ▶ A significant weaknesses to satellite systems is the use of long-range telemetry for communication with ground stations. The uplinks & downlinks are often transmitted through open telecom network security protocols that can be accessed by cyber attackers.
- ▶ Space Policy-Directive 5 urged satellite manufacturers to design their hardware and software so that operators can monitor and adapt to “activities that could manipulate, deny, degrade, disrupt, destroy, surveil, or eavesdrop on space system operations.”



TOP CYBER SECURITY TRENDS



Ransomware Surge: Increase in sophisticated attacks; targets businesses of all sizes and critical infrastructure.

Zero Trust Security: Assumes distrust; enforces strict access controls to combat evolving cyber threats effectively.

Remote Work Security: Ensures secure endpoints, access methods, and data protection due to the global shift to remote work.

AI & ML in Cybersecurity: Utilized for threat detection, anomaly detection, and improving incident response efficiency.

Cloud Security Challenges: Focuses on securing cloud infrastructure, addressing misconfigurations, and preventing data breaches.

IoT & OT Security: Focuses on securing vulnerable Internet of Things (IoT) and operational technology (OT) systems.

Cybersecurity Regulations: Governments enforce new laws, compelling organizations to safeguard sensitive data and critical infrastructure.

Identity & Access Management (IAM): Manages user identities, emphasizes multi-factor authentication, and controls system access.

Supply Chain Vulnerabilities: Involves exploiting vulnerabilities in third-party vendors and suppliers to gain access to target organizations.

Incident Response & Threat Hunting: Prepares for and responds to cyber incidents; emphasizes proactive threat detection and response planning.

Privacy & Data Protection: Ensures compliance with regulations like GDPR and CCPA; focuses on safeguarding customer data and legal adherence.

AI-Powered Cyber Threats: AI used by cybercriminals for sophisticated and targeted attacks, posing evolving threats to cybersecurity.

Cybersecurity Trends and Strategies

1. Evolving threats require proactive strategies
2. Ransomware-as-a-Service (RaaS) prevalence
3. Nation-state APTs target critical sectors
4. Cloud misconfigurations cause breaches
5. Insider threats pose risks
6. Implement Zero Trust Everywhere approach
7. Adopt responsible and secure AI
8. Prepare for Post-Quantum Cryptography
9. Focus on resilience: recovery and continuity
10. Invest in workforce training and education



SECURITY BY DESIGN, DEFENSE IN DEPTH, ZERO TRUST



- Security by design – build agile systems with operational cyber-fusion to be able to monitor recognize, and respond to emerging threats. Install intrusion prevention systems (IPS) or intrusion detection systems (IDS) to monitor malicious activity on your industrial network. Also enable security settings on energy system networks
- Defense-in –Depth; technologies, processes, air-gapping, hardening, Encryption of data flowing from sensors and segmentation of OT and IT.
- Zero Trust is a security model that uses strict identity verification for every person or entity attempting to access network resources, regardless of whether the person or entity is in the office bound by the network perimeter or accessing the network remotely

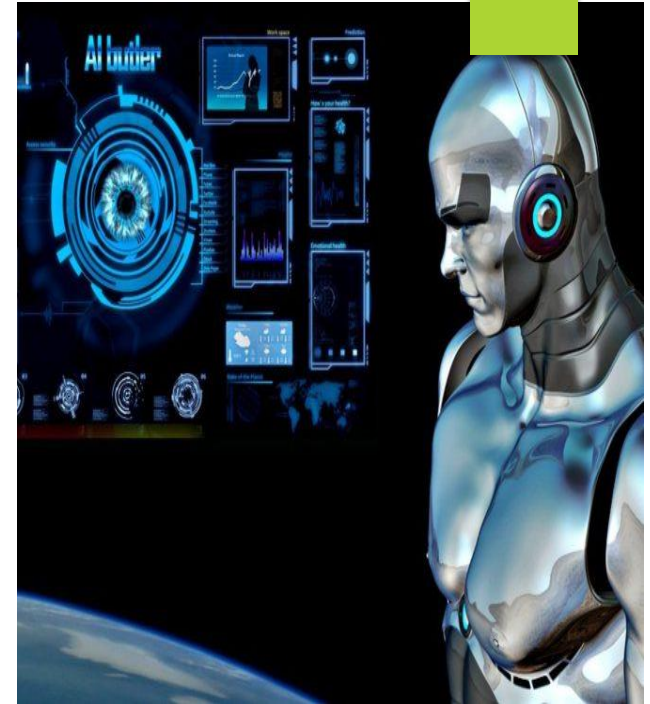
. Essential principles for navigating the technological and cybersecurity future.

- Integrate emerging technologies securely using strategies and frameworks.
- Cybersecurity is everyone's responsibility, from leadership to end-users.
- Protecting Identities and ensuring trust are critical for digital transformation to succeed.



Future technologies will continue to drive innovation and need evolving security measures.

- ▶ 5G/6G + Edge Computing
- ▶ Internet of Things
- ▶ Digital Twins & AR/VR
- ▶ Biometrics
- ▶ Biotechnology + Cyber Convergence: CRISPR, Genetic data, medical IoT devices
- ▶ Nanotechnologies
- ▶ Space Technologies
- ▶ Hypersonics
- ▶ Quantum mechanics
- ▶ Robotics: AI and machine integration.



```
def operation =
```

```
  mirror_and_ase =
```

```
  mirror_and_ase.y
```

```
  mirror_and_ase.x
```

```
  if operation =
```

```
    mirror_and_ase.y
```

```
    mirror_and_ase.x
```

```
    mirror_and_ase.y
```

```
    mirror_and_ase.x
```

```
    # the deselected mirror modifier object
```

```
  mirror_and_ase =
```

```
  mirror_and_ase.y
```

```
  print "Selected" + str(mirror_and_ase) # mirror_and_ase is the active object
```

```
  mirror_and_ase.y
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```
  mirror_and_ase.x
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```
  mirror_and_ase.y
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  mirror_and_ase.x
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Thank You!

- ▶ **Chuck Brooks**
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